

EMCH 260

SOLID MECHANICS

Ming Hu

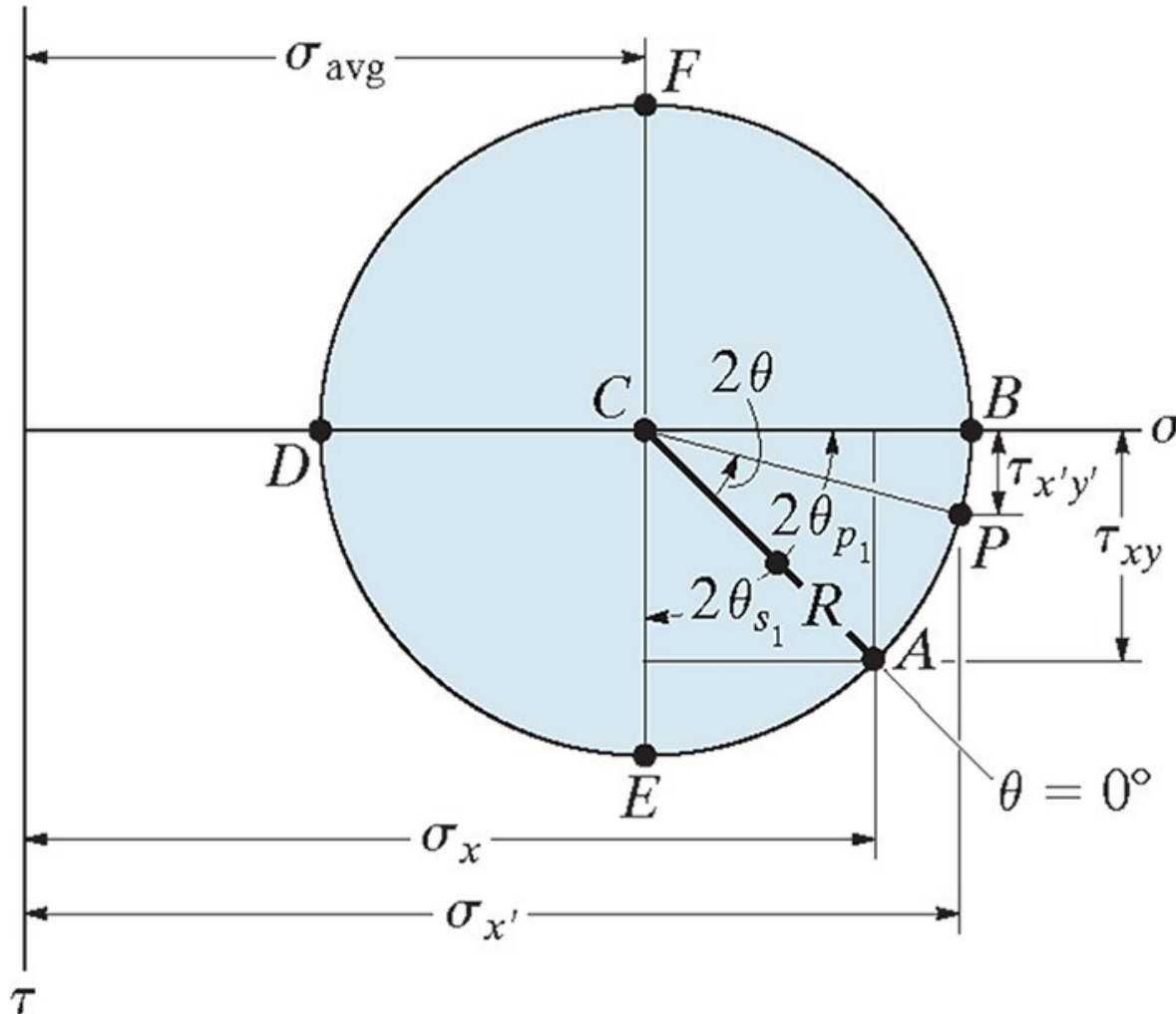
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Stress Transformation

Chapter 9.3

- ✓ Know how to use the Mohr's circles to calculate absolute maximum shear stress

PREVIOUS ON MOHR'S CIRCLE

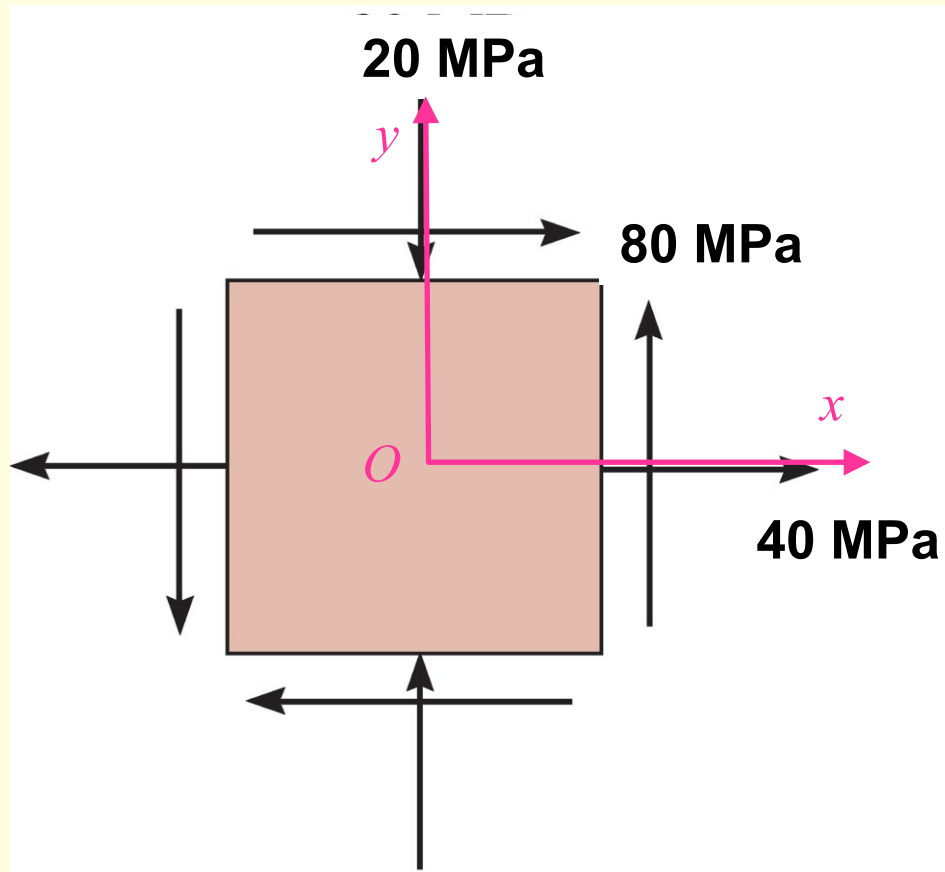


Mohr's Circle:

- Central point **C**
- Radius **R**
- Reference line **CA**
- Principal stresses (**B & D**)
- Maximum in-plane stress (**E or F**)
- State of stress at an arbitrary point (**P**) with orientation relative to original Oxy system (**2θ**)

IN-CLASS EXAMPLE 1

Determine the principal stresses and the maximum in-plane shear stress. Specify the orientation of the element in each case.



IN-CLASS EXAMPLE 1, Cont'd

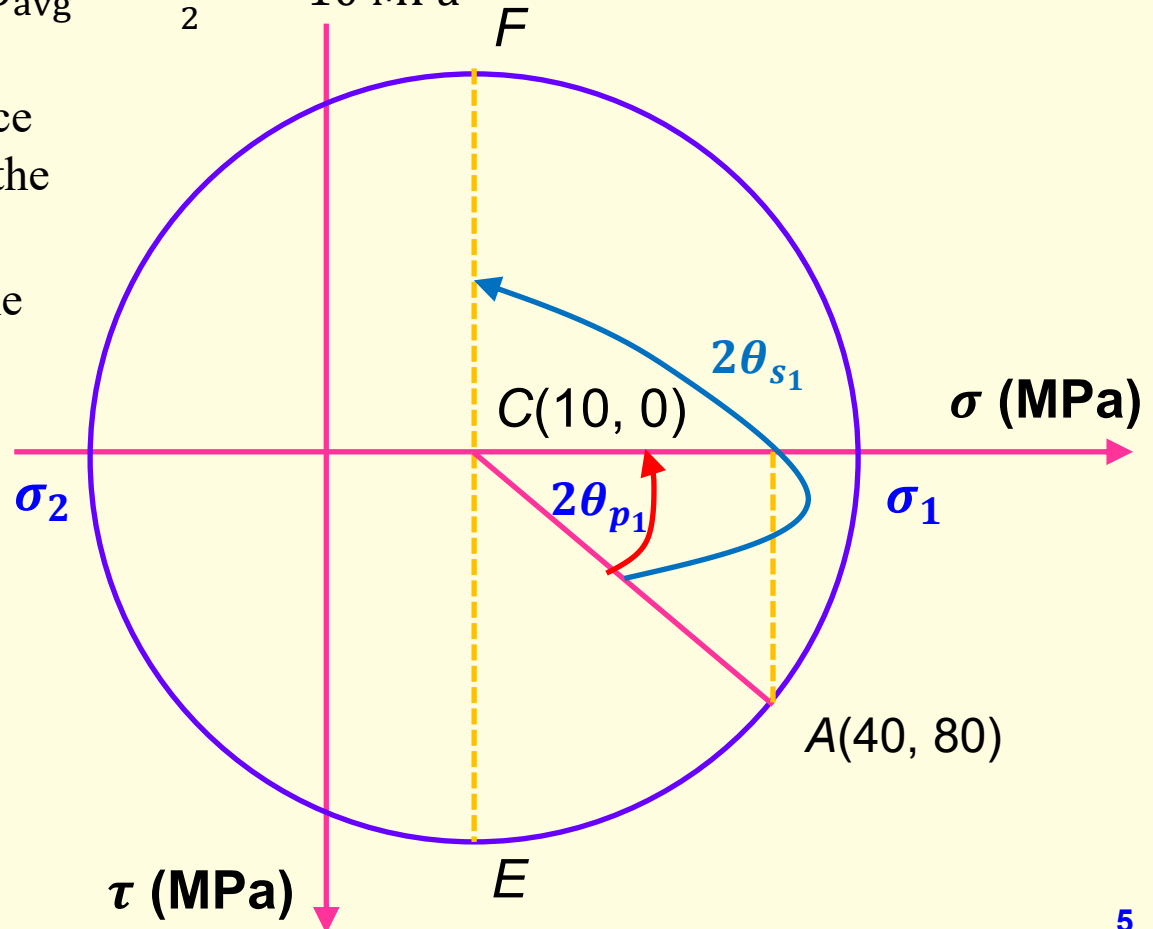
Solutions

Step 1: Construction of the Mohr's Circle

Given $\sigma_x = 40$ MPa, $\sigma_y = -20$ MPa, $\tau_{xy} = 80$ MPa, the center of the circle C is located on the σ -axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 10$ MPa

The point $C(10, 0)$ and the reference point $A(40, 80)$ are labeled. Then, the radius of the circle is $R =$

$\sqrt{(40 - 10)^2 + 80^2} = 85.4$, and the circle is constructed.



IN-CLASS EXAMPLE 1, Cont'd

Solutions, Cont'd

Step 2: Principal Stresses

The principal stresses are the two intersections of the Mohr's circle with the σ -axis

$$\sigma_1 = 10 + 85.4 = 95.4 \text{ MPa}$$

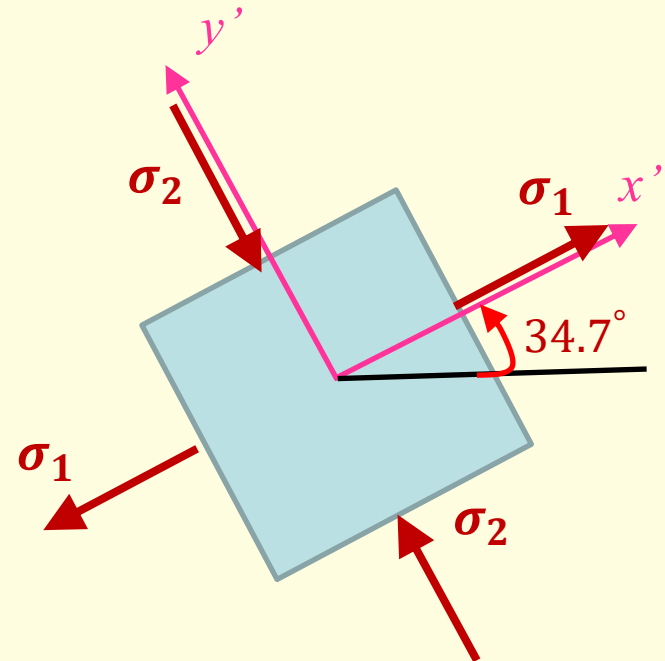
$$\sigma_2 = 10 - 85.4 = -75.4 \text{ MPa}$$

From the Mohr's circle, the counterclockwise angle $2\theta_{p_1}$, measured from CA to the $+\sigma$ axis, is

$$2\theta_{p_1} = \tan^{-1} \left(\frac{80}{40 - 10} \right) = 69.4^\circ$$

$$\text{Thus, } \theta_{p_1} = 34.7^\circ$$

The counterclockwise rotation defines the direction of the x' -axis and σ_1 .



IN-CLASS EXAMPLE 1, Cont'd

Solutions, Cont'd

Step 3: Maximum In-Plane Shear Stress

The maximum in-plane shear stresses are the upper and lower points on the Mohr's circle:

$$\tau_{\max}^{\text{in-plane}} = R = 85.4 \text{ MPa}$$

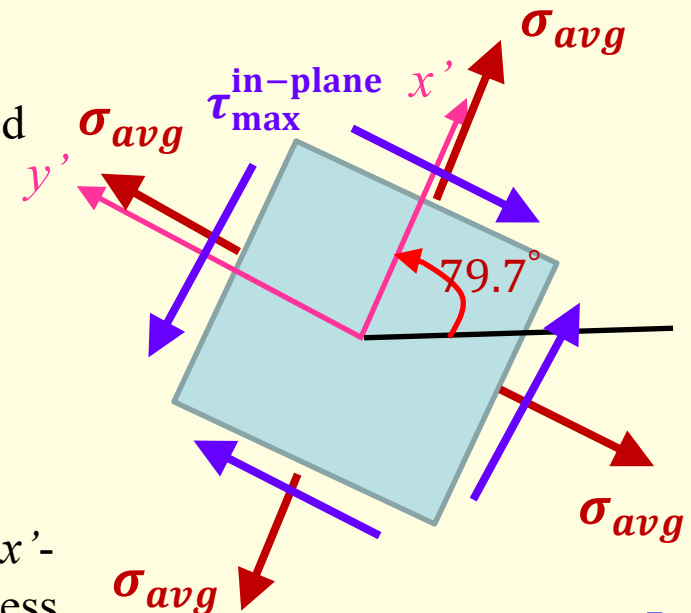
The average normal stress is 10 MPa (the σ -axis value of the center point).

From the circle, the counterclockwise angle $2\theta_{s_1}$, measured from CA to the CF , is

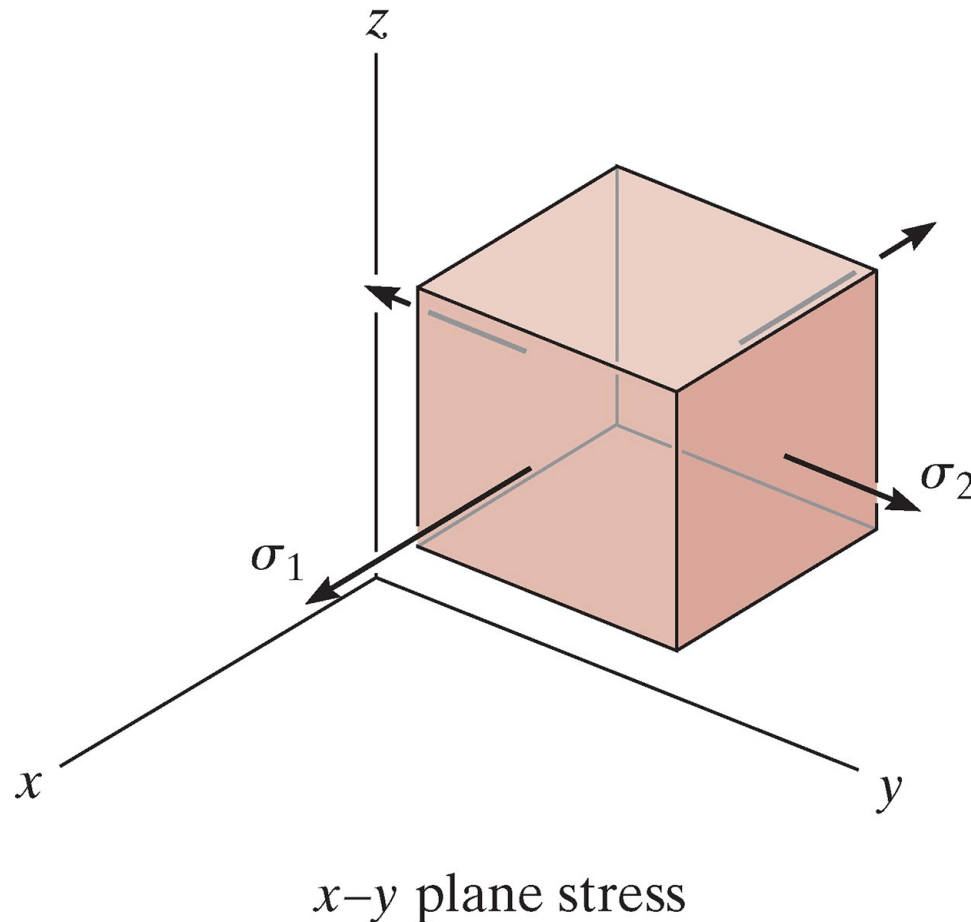
$$2\theta_{s_1} = 2\theta_{p_1} + 90^\circ = 159.4^\circ$$

$$\text{Thus, } \theta_{s_1} = 79.7^\circ$$

The counterclockwise rotation defines the direction of the x' -axis and the negative value of maximum in-plane shear stress.

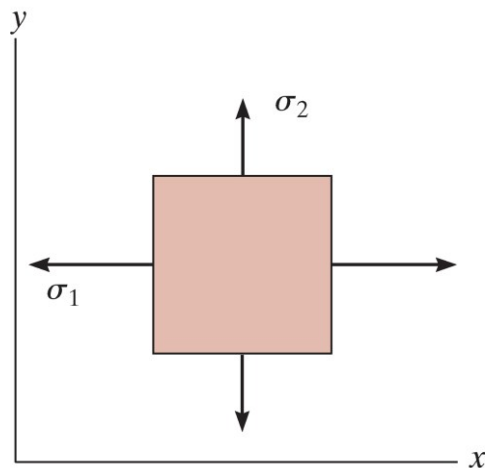
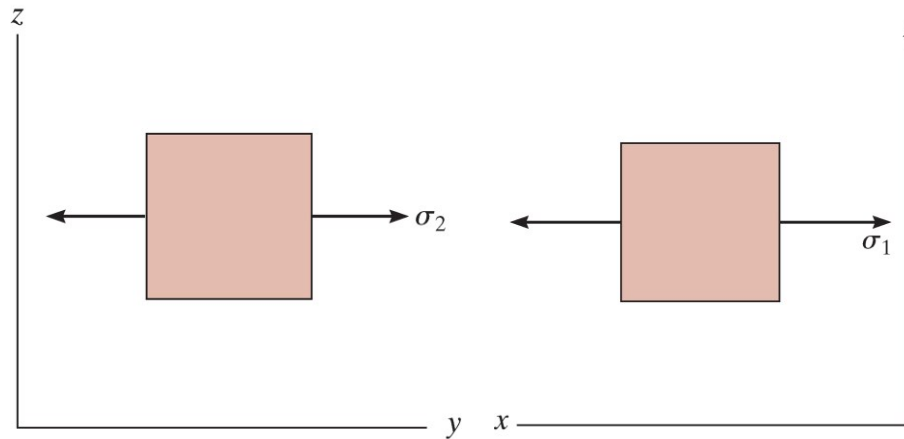


ABSOLUTE MAXIMUM SHEAR STRESS



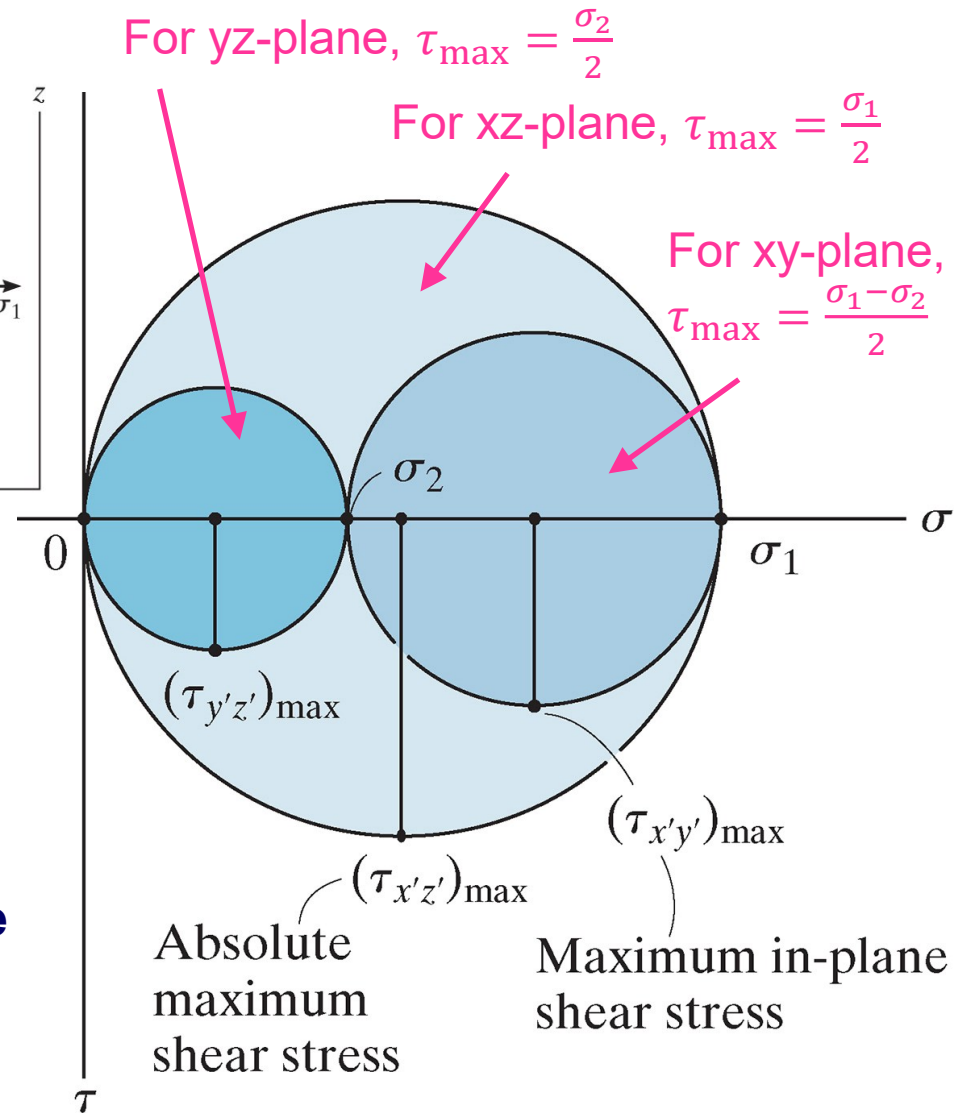
To find out the absolute maximum shear stress in a **3D** material, we can view the element in **2D** at a time, i.e. in the yz , xz , and xy planes, then we use the Mohr's circle to determine the maximum in-plane shear stress for each case.

MOHR'S CIRCLE FOR EACH PLANES

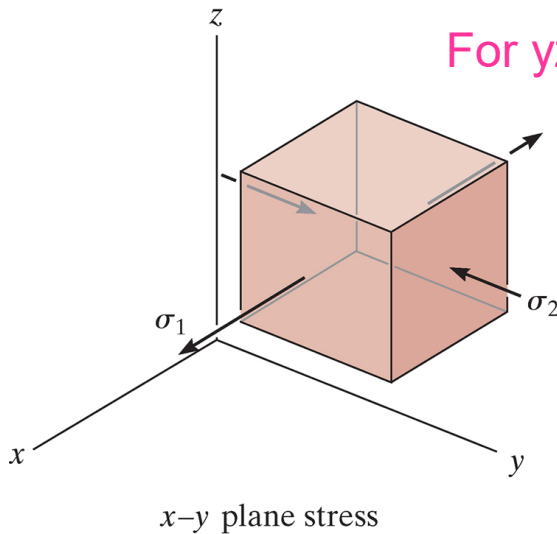


$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2}$$

σ_1 and σ_2 have the same sign and $\sigma_1 > \sigma_2$



WHEN THERE IS COMPRESSIVE STRESS



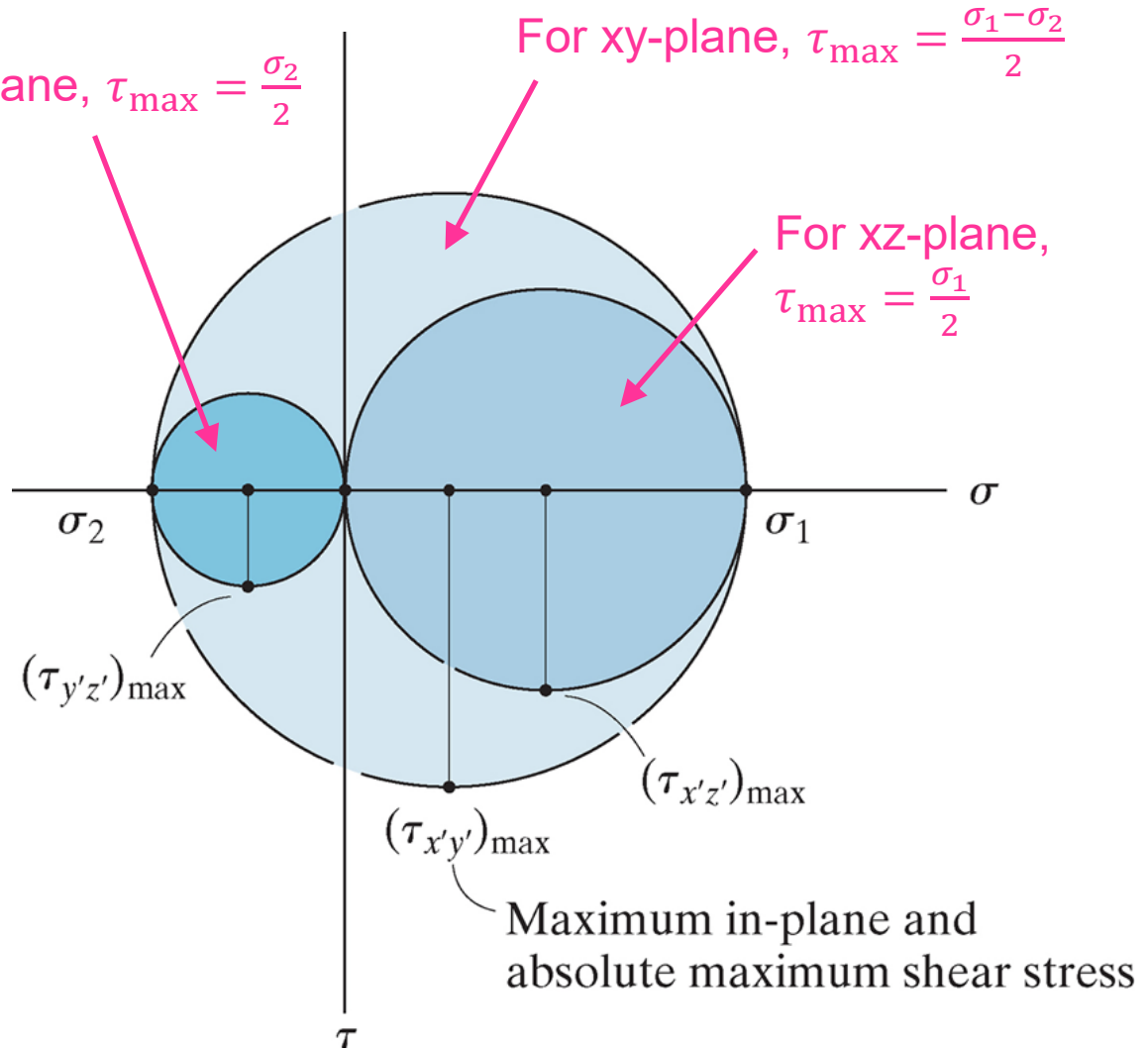
For yz -plane, $\tau_{\max} = \frac{\sigma_2}{2}$

For xy -plane, $\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2}$

For xz -plane, $\tau_{\max} = \frac{\sigma_1}{2}$

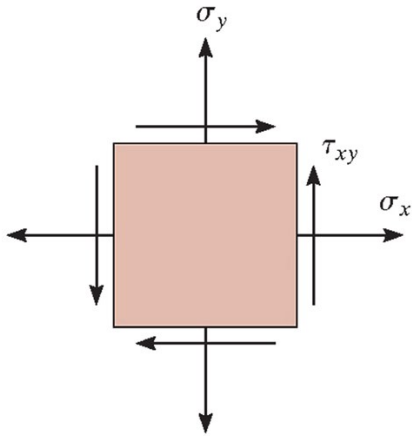
$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2}$$

σ_1 and σ_2 have opposite sign, and $\sigma_1 > 0$, $\sigma_2 < 0$

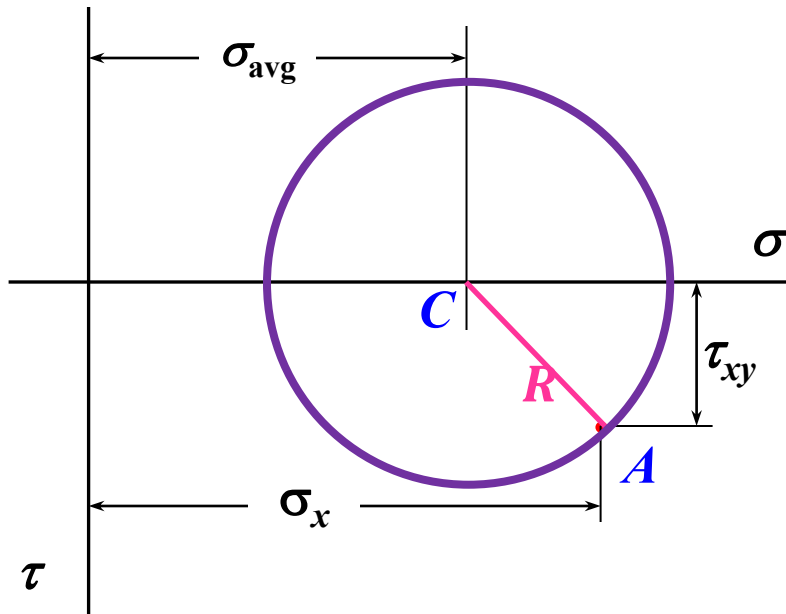


PROCEDURE FOR ANALYSIS

Step 1: Construction of Mohr's circle



A stress state of a point for which all stress components σ_x , σ_y and τ_{xy} are positive (just for example)



- Establish a coordinate system;
 σ - τ axis
- Plot the center of the Mohr's circle
 $C(\sigma_{avg}, 0)$ on σ -axis $\sigma_{avg} = (\sigma_x + \sigma_y)/2$
- Plot the reference point $A(\sigma_x, \tau_{xy})$.
This represents $\theta = 0^\circ$
- Connect point A with the center C ,
and CA becomes the radius of the circle
- Sketch the circle

PROCEDURE FOR ANALYSIS, CONT'D

Step 2: Principal Stresses

With the Mohr's circle, we can calculate the principle stresses and the associated orientation.

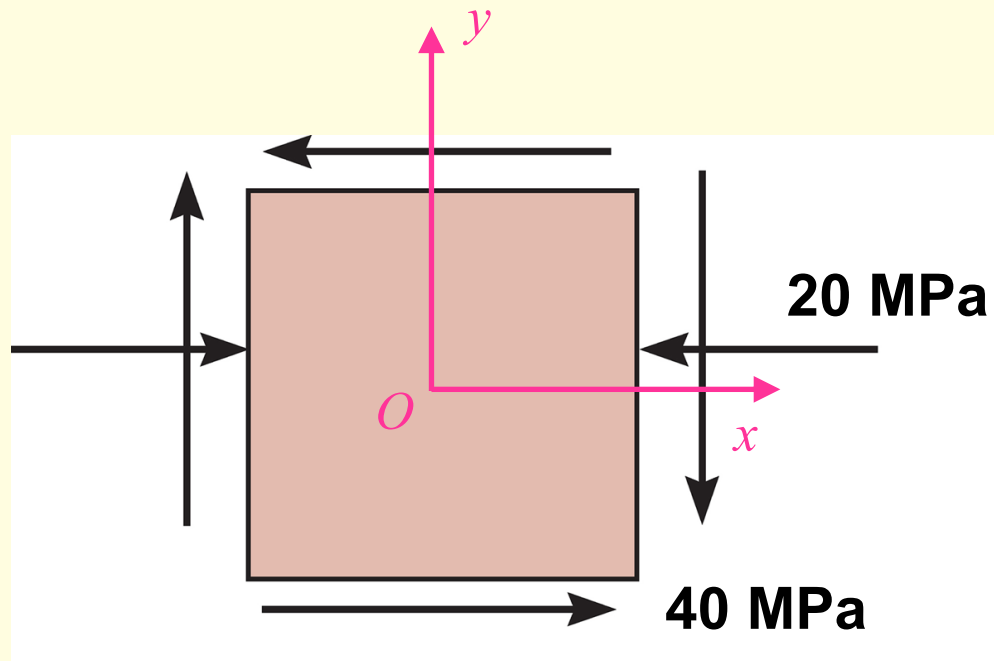
Step 3: Absolute Maximum Shear Stress

Depending on the sign of the obtained principal stresses, the absolute maximum shear stress can be obtained by applying the previous equations.

The same results can be obtained by drawing Mohr's circle for each orientation of an element about the x, y, and z axes, and then comparing the maximum shear stress among the three cases.

IN-CLASS EXAMPLE 2

Due to an applied loading, an element at a point on a machine shaft is subjected to the state of plane stress shown below. Determine the principal stresses and the absolute maximum shear stress at the point.



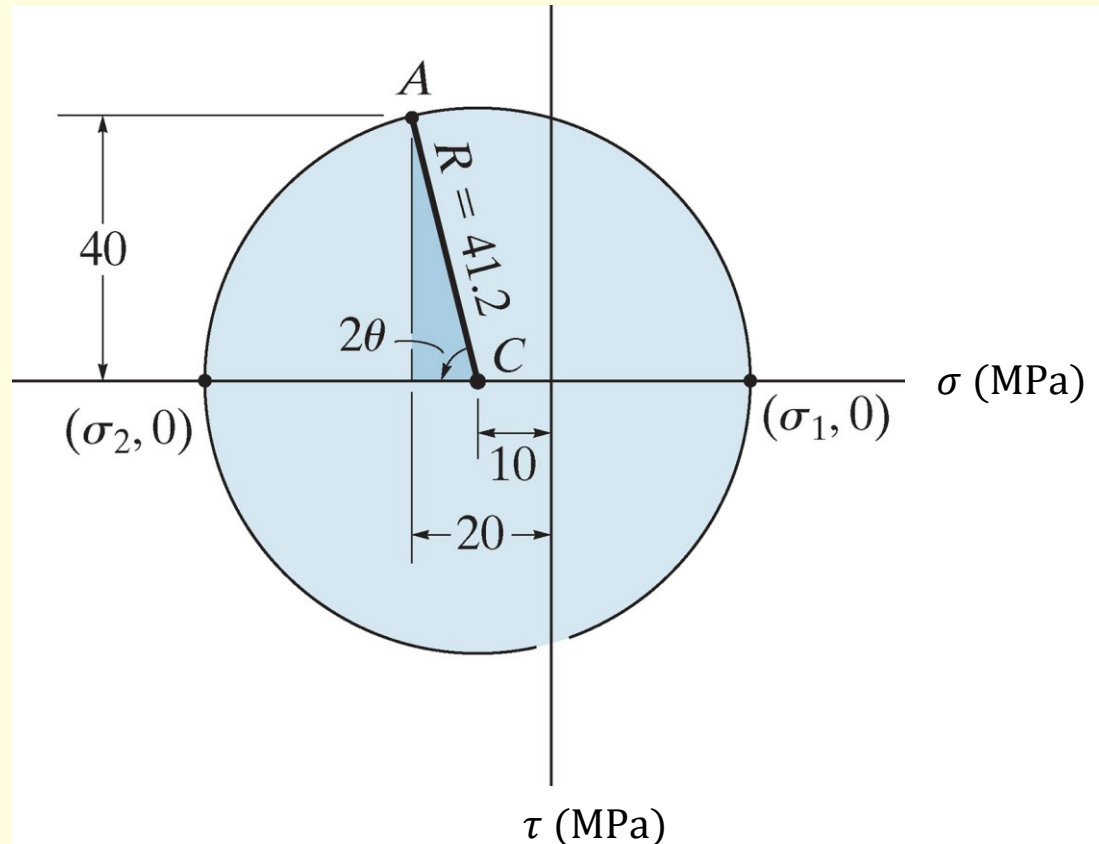
IN-CLASS EXAMPLE 2, Cont'd

Solutions

Step 1: Construction of the Mohr's Circle

Given $\sigma_x = -20$ MPa, $\sigma_y = 0$, $\tau_{xy} = -40$ MPa, the center of the circle C is located on the σ -axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = -10$ MPa

The point $C(-10, 0)$ and the reference point $A(-20, -40)$ are labeled. Then, the radius of the circle is $R = \sqrt{(20 - 10)^2 + 40^2} = 41.2$, and the circle is constructed.



IN-CLASS EXAMPLE 2, Cont'd

Solutions, Cont'd

Step 2: Principal Stresses

$$\sigma_1 = -10 + 41.2 = 31.2 \text{ Mpa}$$

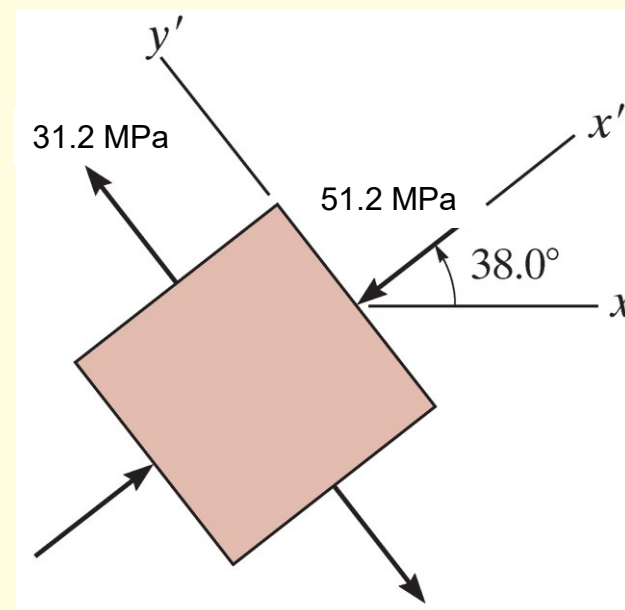
$$\sigma_2 = -10 - 41.2 = -51.2 \text{ Mpa}$$

From the circle, the counterclockwise angle 2θ , measured from CA to the $-\sigma$ axis, is

$$2\theta = \tan^{-1} \left(\frac{40}{20 - 10} \right) = 76.0^\circ$$

$$\text{Thus, } \theta = 38.0^\circ$$

The counterclockwise rotation defines the direction of the x' -axis and σ_2 .



IN-CLASS EXAMPLE 2, Cont'd

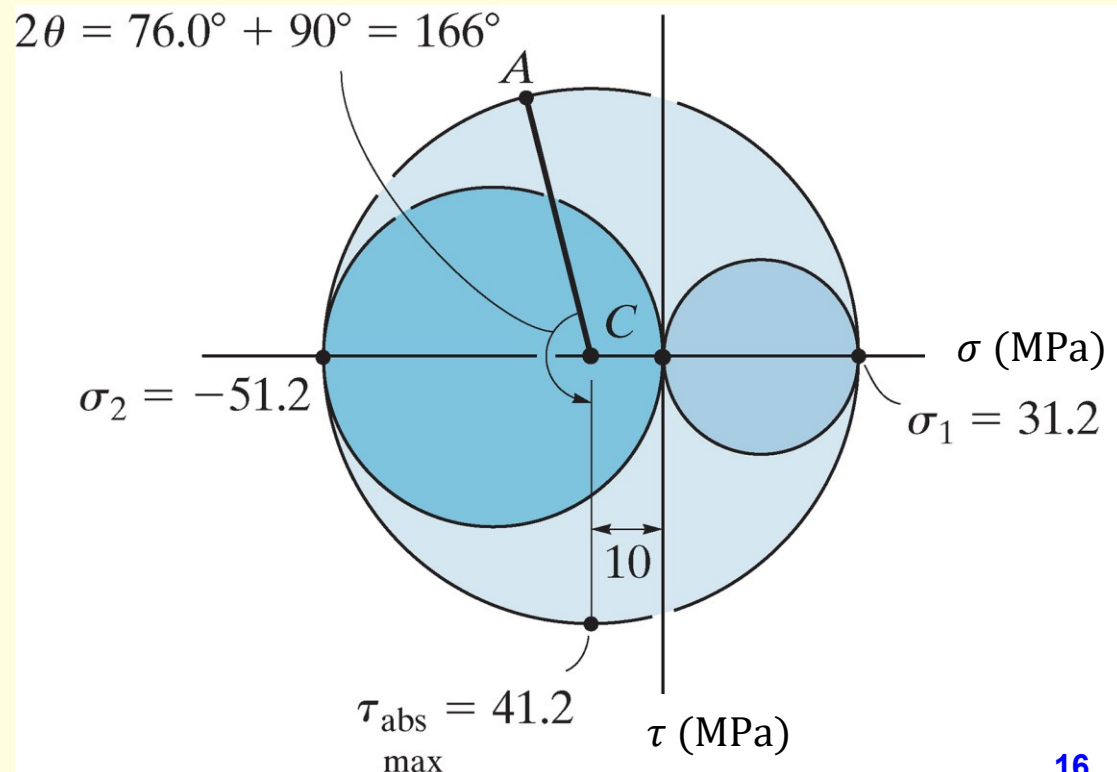
Solutions, Cont'd

Step 3: Absolute Maximum Shear Stress

Since σ_1 and σ_2 have opposite signs, applying Eq. (9-14) on textbook page 482 we have

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = \frac{31.2 - (-51.2)}{2} = 41.2 \text{ MPa}$$

The same results can be obtained by drawing three Mohr's circle for each orientation of an element about the x , y , and z axes, and then comparing the maximum shear stress among the three cases (right figure).



RELEVANT PRACTICE

- **Assignment 3 on Blackboard, problem #4 (go to Blackboard under the “Assignments” folder, **Assignment 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**
- **Optional Extra Exercise 3 on Blackboard, problem #4 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 481 – 484**
 - **In particular, Examples 9.10 – 9.11**